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Projekt Documentation
Modeling Rally Entries with RDF and Wikidata

1. Project aim

This project defines an RDF/RDFS model of rally events and crew participation. It describes the final podium of the 1999 Monte Carlo Rally and two selected special stages. The model can answer questions about the participants, entrant teams, car specifications, results and described stages. The project continues an earlier assignment that queried rally information in Wikidata with SPARQL. It introduces local classes and properties and links selected resources to Wikidata.

2. Overall graph structure

The graph combines a vocabulary with example data. The vocabulary defines classes and properties and uses RDFS domains and ranges to support type inference. The model applies this vocabulary to actual data using named resources such as WRC, WRC1999, and MonteCarlo1999. The championship context starts with WRC, a ChampionshipSeries. hasSeason connects it to WRC1999, a ChampionshipSeason, and hasEvent connects that season to MonteCarlo1999, a RallyEvent. This separates the competition across years, one particular season and one event. seasonYear belongs to the season, while the event carries its dates, round number and distance values.

The event also has a separate rally context. MonteCarlo1999 is connected by editionOf to MonteCarloRally, a RallySeries. MonteCarloRally identifies the recurring rally, while MonteCarlo1999 identifies its 1999 occurrence. inSeason places that occurrence in the championship calendar. These two links describe different relationships.

RallyEntry connects participation to this event structure. MonteCarlo1999 hasEntry EntryMakinen1999. That entry hasDriver TommiMakinen, hasCoDriver RistoMannisenmaki, representsTeam MarlboroMitsubishiRalliart and usesCarModel MitsubishiLancerEvoVI. Its finishingPosition is 1. Together, these statements describe one crew and car participating for one team in one event. They do not give a person a permanent team, car or finishing position.

Other information is attached to the resource it describes. An event hasStage links to stage occurrences, each with its own stageNumber and stageLengthKm. hasSurface connects the event to surface or condition categories. People, teams, car specifications and surfaces have their own identifiers and can be reused by further entries or events.

A simpler alternative would attach the driver, team, car and result directly to the event. With several crews, that would leave the combinations unclear. The graph would not identify which co-driver, team and car belong together. Attaching them to a driver would also lose the event context. A separate RallyEntry groups those relationships and provides the event context.

3. Class dictionary and taxonomy

Driver and CoDriver are subclasses of Person through rdfs:subClassOf. A person can perform different roles in different events. The role in a particular entry is identified by hasDriver or hasCoDriver.

Person. A human who can act as a driver or a co-driver.

Driver. A person occupying the driver role in a rally entry.

CoDriver. A person occupying the navigator/co-driver role in a rally entry.

ChampionshipSeries. A recurring multi-season competition such as the World Rally Championship.

ChampionshipSeason. One yearly edition of a championship series.

RallySeries. A recurring rally such as the Monte Carlo Rally, not a particular annual edition.

RallyEvent. A particular edition of a rally, held on specific dates.

RallyEntry. One crew and car-model entry in one rally event. It records event-specific team affiliation and finishing position.

Team. An entrant/competition team associated with a rally entry.

CarModel. A rally car model or competition specification. It does not denote a unique physical vehicle.

RallyStage. An individual timed special stage belonging to one rally event.

Surface. A surface/condition category relevant to a rally event, such as asphalt or snow.

The championship and rally contexts connect instances through ordinary properties. An annual season or event is not represented as a subclass of its recurring competition.

Classes use rdfs:Class, properties use rdf:Property, and instances use rdf:Description with explicit rdf:type statements. Driver and CoDriver are subclasses of Person. hasDriver is a relationship between an entry and a person, not a subclass relationship.

The local namespace is https://example.org/rally#. rdf:about identifies the described resource, rdf:resource refers to another resource. Dates, numbers and names are literals.

4. Properties linking resources

These fourteen properties connect one resource to another. All are declared as rdf:Property. Each entry lists its domain, range and intended meaning.

hasSeason. Domain: ChampionshipSeries. Range: ChampionshipSeason. A championship series includes this season.

seasonOf. Domain: ChampionshipSeason. Range: ChampionshipSeries. The parent recurring championship of a championship season.

hasEvent. Domain: ChampionshipSeason. Range: RallyEvent. A championship season includes a rally event.

inSeason. Domain: RallyEvent. Range: ChampionshipSeason. The championship season in which an event was held.

editionOf. Domain: RallyEvent. Range: RallySeries. A rally event is an annual edition of a recurring rally series.

hasEntry. Domain: RallyEvent. Range: RallyEntry. The rally event has this specific crew entry.

entryForEvent. Domain: RallyEntry. Range: RallyEvent. The specific rally event for which the entry was registered.

hasDriver. Domain: RallyEntry. Range: Driver. The driver role filled for this particular entry.

hasCoDriver. Domain: RallyEntry. Range: CoDriver. The co-driver role filled for this particular entry.

representsTeam. Domain: RallyEntry. Range: Team. The team under which an entry competed in the specific rally.

usesCarModel. Domain: RallyEntry. Range: CarModel. The model/specification used in this entry; not a unique chassis.

hasStage. Domain: RallyEvent. Range: RallyStage. A rally event comprises a timed special stage.

stageOfEvent. Domain: RallyStage. Range: RallyEvent. The event that contains this timed special stage.

hasSurface. Domain: RallyEvent. Range: Surface. A rally event features the listed road surface/condition.

5. Properties with literal values

These eleven properties have date or numerical values. All distances are measured in kilometres. xsd refers to http://www.w3.org/2001/XMLSchema#.

seasonYear. Domain: ChampionshipSeason. Range: xsd:gYear. Calendar year of a championship season.

startDate. Domain: RallyEvent. Range: xsd:date. First calendar day of the entire rally event, including any opening noncompetitive liaison leg.

endDate. Domain: RallyEvent. Range: xsd:date. Last calendar day of an individual rally event.

roundNumber. Domain: RallyEvent. Range: xsd:positiveInteger. Ordinal round within the championship season.

editionNumber. Domain: RallyEvent. Range: xsd:positiveInteger. Edition number of the recurring rally.

numberOfStages. Domain: RallyEvent. Range: xsd:nonNegativeInteger. Full number of timed stages in the event (not number currently described in this sample).

competitiveDistanceKm. Domain: RallyEvent. Range: xsd:decimal. Sum of timed special-stage distances in kilometres, not the entire route.

totalRouteDistanceKm. Domain: RallyEvent. Range: xsd:decimal. All route kilometres including noncompetitive liaison sections.

finishingPosition. Domain: RallyEntry. Range: xsd:positiveInteger. Official final classification rank of a crew entry.

stageNumber. Domain: RallyStage. Range: xsd:positiveInteger. Ordinal number of a special stage in the rally.

stageLengthKm. Domain: RallyStage. Range: xsd:decimal. Timed length of one special stage in kilometres.

6. Sample data and modeling decisions

MonteCarlo1999 is the opening round of the 1999 World Rally Championship, the 67th edition of the recurring rally. Its dates are 17–20 January 1999. The start includes the opening noncompetitive liaison leg from Monaco to Gap, timed special stages began on 18 January. competitiveDistanceKm is 424.69, while totalRouteDistanceKm is 1613.20 and includes liaison sections. numberOfStages records the full event total of fourteen. The graph describes only SS01, Plan de Vitrolles – Faye, at 48.28 km, and SS10, Lantosque – Lucéram 1, at 20.87 km.

EntryMakinen1999 combines Tommi Mäkinen, Risto Mannisenmäki, Marlboro Mitsubishi Ralliart and Mitsubishi Lancer Evolution VI, with finishingPosition 1. EntryKankkunen1999 combines Juha Kankkunen, Juha Repo, Subaru World Rally Team and Subaru Impreza S5 WRC 99, with position 2. EntryAuriol1999 combines Didier Auriol, Denis Giraudet, Toyota Castrol Team and Toyota Corolla WRC, with position 3.

RallyEntry groups these facts within one event context. Attaching a team, car or result directly to a driver would leave the event unspecified and could mix information from different participations. Additional events can reuse the same people and car specifications through new entries. The present sample contains one event and therefore does not demonstrate a team or car change.

CarModel represents a competition specification rather than a physical chassis. The team resources use event entrant names, they do not establish permanent employment or legal corporate identity. Stage resources identify timed occurrences within this event. Asphalt and Snow are event surface or condition categories and do not specify where or when each condition occurred.

7. Wikidata references

Nine rdfs:seeAlso links point to Wikidata. WRC to Q7852, WRC1999 to Q839687, MonteCarloRally to Q725531, MonteCarlo1999 to Q5646935, TommiMakinen to Q217260, RistoMannisenmaki to Q905647, JuhaKankkunen to Q313708, DidierAuriol to Q313387, and MitsubishiLancerEvoVI to Q44591. rdfs:seeAlso indicates a resource that may provide additional information. It does not formally assert identity or merge local and external descriptions.

8. Discussion: Reusability, Strengths, and Limitations

Domain Benefits and Reusability

This RDF model provides a standardized approach to structuring motorsport data, which is traditionally fragmented across historical archives, independent websites, and unstructured documents. By defining a clear, machine-readable vocabulary for championships, events, and entries, this graph can serve as a foundational knowledge base for external applications. For example, sports journalists or statisticians could use a scaled version of this graph to query complex historical head-to-head statistics across multiple decades. Video game developers or simulation platforms could reuse the structured data to accurately recreate historical rally seasons, linking drivers to their exact car specifications and teams for a given year. Furthermore, the inclusion of rdfs:seeAlso links to Wikidata connects this specialized rally ontology with the broader Semantic Web.

Strengths of the Data Model

The primary strength of this modeling approach is the implementation of the RallyEntry class. In motorsport, drivers frequently change co-drivers, teams, and car specifications not just between seasons, but even between individual events. By avoiding direct links between a driver and a car, and instead grouping them within a specific event context via RallyEntry, the model successfully prevents data entanglement and historical inaccuracies. Additionally, the strict use of XSD datatypes ensures that the graph can be reliably queried using SPARQL numerical filters.

Limitations and Future Work

While effective for overall event results, the current RDFS model has limitations in capturing dynamic, granular race data. It models static final classifications but does not currently account for temporal in-race events such as stage split times, time penalties, or mechanical retirements. Representing these would require a more complex temporal ontology. Furthermore, relying purely on RDFS limits the strictness of the model. Broader domain reuse would benefit from migrating to the Web Ontology Language (OWL) to enforce disjointness (ensuring a CarModel cannot be classified as a Person) and automated inverse property reasoning, which would further solidify the knowledge base for third-party applications.